



Mariano Zunino

BACKEND ENGINEER / TECHNICAL LEAD - AI & DISTRIBUTED SYSTEMS

📍Montevideo, Uruguay • ✉️marianoz@posteo.net • 🌐mzunino.com.uy
🔗marianozunino • 📄mariano-z

Backend engineer and technical lead focused on scalable backend services, Java/Spring/Micronaut and TypeScript systems, distributed architecture, and AI-powered product features.

Work Experience

STUZO & PAR

Software Architect

Dec. 2023 – Present

- Delivered hands-on backend work across multiple services using Nest.js, TypeScript, GraphQL, and PostgreSQL.
- Built AI-powered features and conversational agents with the Vercel AI SDK and Databricks (Genie / AgentBricks), including SSE streaming endpoints and a multi-agent evaluation harness with MLflow.
- Evaluated feature feasibility and shaped technical designs, helping streamline delivery and improve product quality.
- Partnered with engineering, product, and business stakeholders to align technical solutions with defined scope and priorities.

Software Engineer

Jan. 2023 – Dec. 2023

- Built and maintained backend services with Nest.js, TypeScript, GraphQL, Kubernetes, and AWS.

VAIRIX

Senior Backend Engineer

May 2021 – Jan. 2023

- Built backend services on Seekr using Java, Spring, Micronaut, and Kafka, contributing to distributed event-driven systems.
- Acted as Tech Lead and Architect for PoliticalIQ (Nest.js, TypeScript, AWS).
- Contributed to backend development on Vooks using Node.js, AWS Lambda, and Kubernetes.
- Led technical interviews and mentored backend developers.

NUBCEO

SSr. Backend Developer

Feb. 2020 – May 2021

- Designed microservices for payment data aggregation using Node.js, TypeScript, MongoDB, and AWS.

ANTEL

Software Developer

Apr. 2019 – Feb. 2020

- Developed adapters to integrate heterogeneous systems within the company.
- Managed Kubernetes deployments to ensure scalable and reliable service delivery.

IT Support / Admin

Oct. 2010 – Apr. 2019

- Maintained Linux OS images and provided full-stack IT support.
- Administered OCS systems and improved organizational productivity.

Teaching & Consulting

UNIVERSIDAD DEL TRABAJO DEL URUGUAY (PART-TIME)

Montevideo, Uruguay

University Professor

Aug. 2023 – Present

- Teach “Programming Principles” in C, with emphasis on imperative programming, algorithms, basic data structures, and problem-solving.
- Deliver “Data Structures and Algorithms” using C, focusing on complexity analysis, data structure design, and search/sort techniques.

- Serve as an instructor in the “Information Systems Workshop (.NET)” on designing and developing medium-to-large enterprise systems on the Microsoft .NET platform, emphasizing architecture, teamwork, self-directed learning, and technical presentations.

GRUPO ELIS (PART-TIME)

Architect

May 2024 – Present

- Redesigned and migrated a system for conference and event registration.
- Modernized technical stack using Golang, TypeScript, Node.js, PostgreSQL in a monorepo architecture.

HACK ACADEMY (PART-TIME)

Montevideo, Uruguay

Sep. 2021 – Jan. 2023

- Taught Node.js to students with varying levels of programming knowledge.
- Offered hands-on mentoring to help learners master core backend concepts.

NUBCEO (PART-TIME)

Technical Consultant

May 2021 – 2023

- Provided architectural guidance, conducted code reviews, and led technical evaluations.

Education

TECNÓLOGO EN INFORMÁTICA

Facultad de Ingeniería, Universidad de la República

2018 – 2021

- Focused on software development and IT systems design.

BACHELOR IN IT

Instituto Tecnológico Industrial (UTU)

2008 – 2011

- Studied foundational IT concepts and practical software applications.

C1 ADVANCED CERTIFICATE

Anglo Institute

2024 – 2024

- Achieved CEFR C1 level in English proficiency (Cambridge Advanced Certificate).

Skills

 **Programming Languages**

Java / TypeScript / JavaScript • Go / C# / Python • GraphQL • REST / gRPC / WebSockets

 **AI & LLM Tooling**

Agentic AI / conversational agents • Vercel AI SDK • Databricks (Genie / AgentBricks) • Multi-agent systems / SSE streaming • MLflow evaluation

 **Backend Frameworks & Databases**

Spring / Micronaut / Nest.js / Express / .NET Core • PostgreSQL / MongoDB / Redis / Neo4j

 **Cloud Platforms & DevOps**

AWS • Docker / Kubernetes • Terraform • CI/CD pipelines • Git / Linux / Shell scripting

 **Messaging & Queues**

Kafka / RabbitMQ / ActiveMQ / SQS

 **Languages**

Spanish (Native) • English (C1 – Advanced)